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Name: Harsh kumar Saw Subject: Physiology Homework

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School/College:

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Action Potential

Definition

An action potential is a rapid, temporary change in the membrane potential of an excitable cell, such as a nerve or muscle cell, when an adequate stimulus is applied.

⇒ It follows the all-or-none law.

Phases of action potential

1. Resting phase

- Membrane is polarized
- Inside is negative relative to outside
- Neuronal resting membrane potential is -70 mV

2. Depolarization

- Voltage-gated Na^+ channels open.
- Na^+ enters the cell rapidly.
- Membrane potential becomes positive inside.

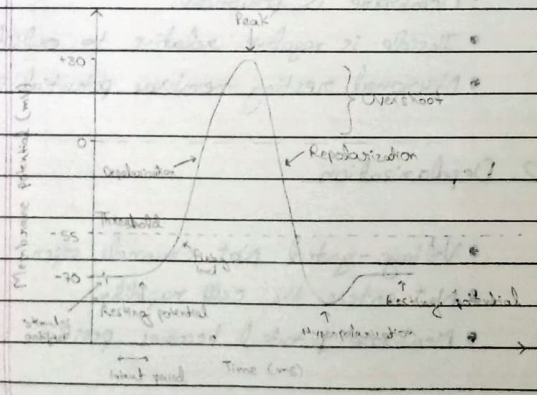
3. Repolarization

- Na^+ channels become inactivated
- Voltage-gated K^+ channels open.
- K^+ leaves the cell, making the inside -ve again.

4. After-hyperpolarization

- K^+ channels close slowly.
- Membrane potential temporarily becomes more negative than the resting level.
- The membrane then returns to RMP.

Action Potential Curve



Important properties

- All-or-none response: Once threshold is reached, a full action potential occurs.
- Threshold: Minimum membrane depolarization required to trigger an action potential.
- Refractory period: Period during which another action potential cannot be generated normally or requires a stronger stimulus.
- Propagation: The action potential travels along the nerve or muscle membrane without decrement in a healthy, excitable membrane.
- Ion movement: Na^+ influx causes depolarization, K^+ efflux causes repolarization.

Clinical importance: Action potentials are essential for nerve impulse transmission, skeletal muscle contraction, and cardiac electrical activity.

Genesis of Resting Membrane Potential (RMP)

Definition

Resting membrane potential (RMP) is the electrical potential difference across the cell membrane at rest, with the inside of the cell negative relative to the outside.

- Neuron: approximately −70 mV
- Skeletal muscle fibre: approximately −90 mV

Mechanism of genesis of RMP

RMP develops due to the unequal distribution of ions across the cell membrane and its selective permeability to different ions.

1. Unequal distribution of ions

- Na^+ concentration is higher in extracellular fluid
- K^+ concentration is higher in intracellular fluid.
- Negatively charged proteins and other anions are present inside the cell.

2. Selective permeability of the membrane

- At rest, the membrane is much more permeable to K^+ than to Na^+.
- K^+ diffuses out of the cell through potassium leak channels.
- This leaves behind negatively charged intracellular substances, making the inside negative.

3. Electrical gradient develops

- As K^+ leaves the cell, the inside becomes increasingly negative.
- This negative charge attracts K^+ back into cell
- Eventually, the electrical force oppose the outward diffusion of K^+.

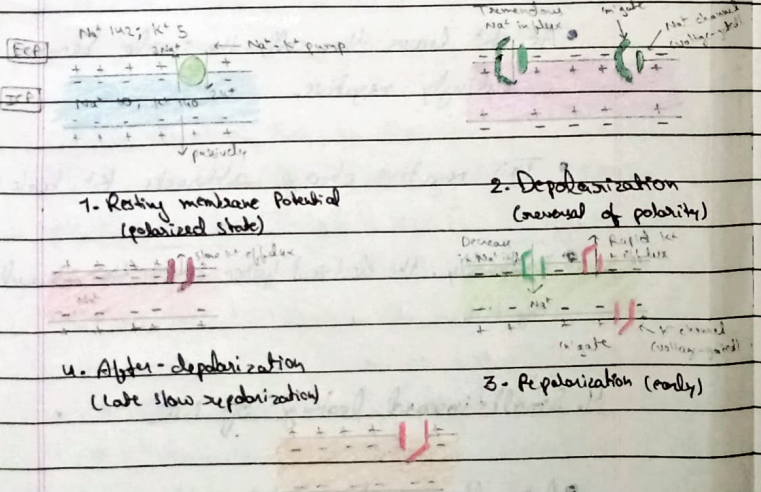
4. Small inward leakage of Na^+

- A small amount of Na^+ enters the cell through leak channels.

- This slightly reduces the negativity produced by K^+ movement.

5. Role of the sodium-potassium pump

- The Na^+/K^+ - ATPase uses one ATP molecule to transport 3 Na^+ out and 2 K^+ into the cell.
- It maintains the ionic concentration gradients
- Because more positive charges leave than enter, it also contributes slightly to the negative RMP.



Development of WBCs (Leukopoiesis)

Definition

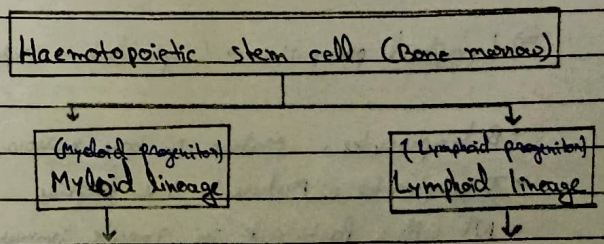
Leukopoiesis is the process of formation and development of white blood cells.

⇒ It occurs mainly in the bone marrow. T lymphocytes undergo further maturation in the thymus.

Types of WBCs

WBCs are divided into two main groups:

- Granulocytes: Neutrophils, eosinophils and basophils
- Agranulocytes: Lymphocytes and monocytes



Development of individual WBCs

A. Granulopoiesis

⇒ Development of granulocyte in the bone marrow.

⇒ Myeloblast → Promyelocyte → Myelocyte → Metamyelocyte → Band cell → Mature granulocyte

⇒ This process produces neutrophils, eosinophils and basophils

B. Monopoiesis

⇒ Monoblast → Promonocyte → Monocyte → Tissue macrophage

⇒ Monocytes circulate in blood and can differentiate into macrophages in tissue.

C. Lymphopoiesis

⇒ Lymphoid progenitor → Lymphoblast → Mature lymphocyte

- B lymphocytes: Mature in bone marrow in humans
- T lymphocytes: Mature in the thymus
- NK cells: Participate in innate immune defence

Functions of WBCs

- Neutrophils: Phagocytosis of bacteria; acute inflammation
- Eosinophils: Defence against parasites; allergic response
- Basophils: Release histamine and other inflammatory mediators
- Monocytes: Become macrophages; phagocytosis
- Lymphocytes: Adaptive immunity and immune surveillance

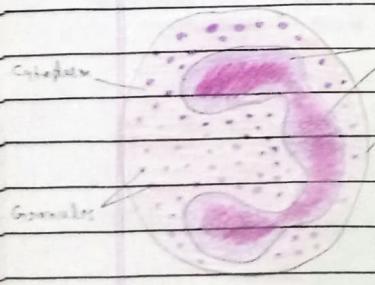
Clinical importance: Changes in WBC count and differential count help in assessing infections, allergic disorders, inflammation and haematological diseases.

Granulocytes and Agranulocytes

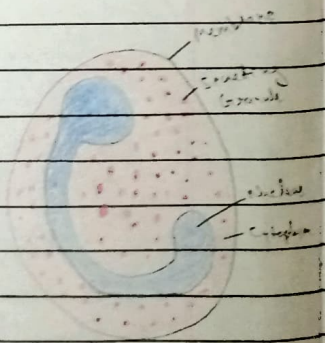
Classification of WBCs

WBCs are nucleated blood cells involved in body defence and immunity. Normal total WBC count in adults is approximately 4,000 - 11,000 cells/μL

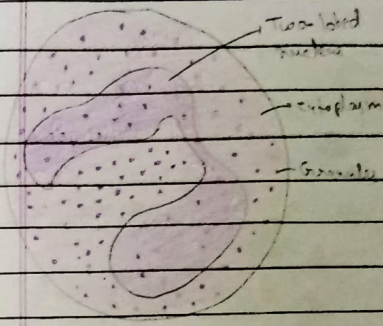
A. Granulocytes



1. Neutrophil

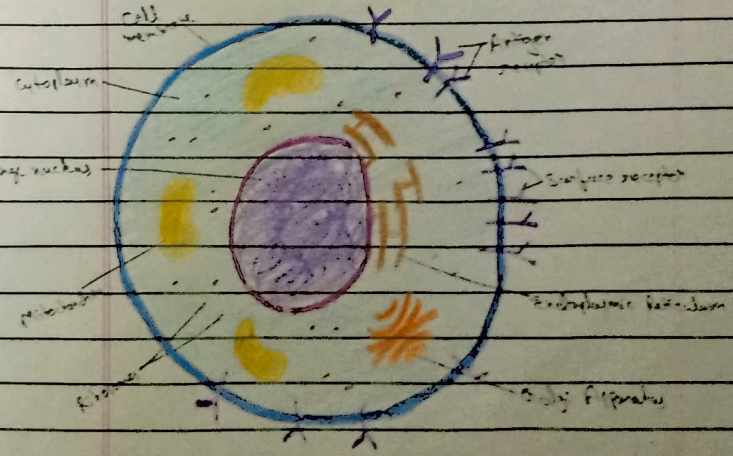


2. Eosinophil

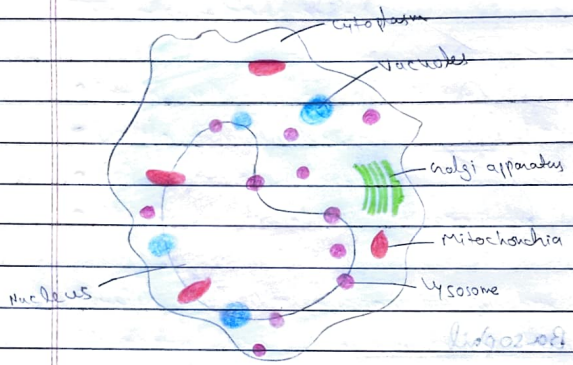


3 Basophil

B. A Granulocytes



4. Lymphocyte



5. Monocyte

Paralysis

Definition

Paralysis is the loss of voluntary muscle activity due to failure of the motor pathway, neuromuscular transmission or skeletal muscle function.

Physiological Basis

Normal voluntary muscle contraction requires:

1. Generation of an action potential in the motor neuron.
2. Conduction of the nerve impulse along the motor nerve.
3. Release of acetylcholine at the neuromuscular junction.
4. Binding of acetylcholine to receptors on the motor end plate.
5. Generation of a muscle action potential.
6. Excitation-contraction coupling and interaction of actin and myosin.

Failure of any of these processes may result in weakness or paralysis.

Mechanism of Paralysis

1. Paralysis due to nerve injury

Damage to a motor nerve interrupts the conduction of impulses to skeletal muscle.

This results in:

- Loss of voluntary muscle contraction
- Decreased muscle tone.
- Reduced or absent reflexes.
- Muscle wasting in prolonged denervation

2. Paralysis due to neuromuscular junction disorder.
Neuromuscular transmission occurs through the release of acetylcholine from the motor nerve terminal.

Disorders affecting acetylcholine release or its receptors impair transmission of nerve impulses to skeletal muscle.

Myasthenia gravis: An autoimmune disorder in which antibodies act against acetylcholine receptor at the neuromuscular junction. This reduces neuromuscular transmission and produces muscle weakness and fatigability

Botulism: Botulinum toxin inhibits the release of acetylcholine from the parasynaptic nerve terminal, resulting in flaccid paralysis

3. Paralysis due to neuromuscular blocking agents
Neuromuscular blocking agents interfere with transmission at the neuromuscular junction.

They may act by:

- Blocking acetylcholine receptors
- Producing persistent depolarization of the motor end plate

This prevents effective muscle action potential and produces skeletal muscle paralysis.

These agents are used clinically during general anaesthesia to facilitate endotracheal intubation and muscle relaxation.

4. Paralysis due to muscle disorders
Diseases affecting skeletal muscle may impair excitation-contraction coupling or the contractile mechanism.

Example include muscular dystrophy and other myopathies

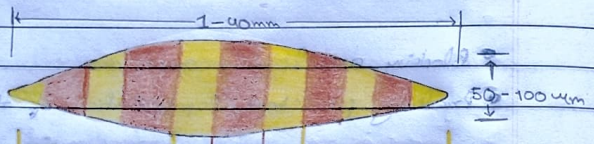
These disorders produce muscle weakness and may cause severe loss of muscle function.

Conclusion

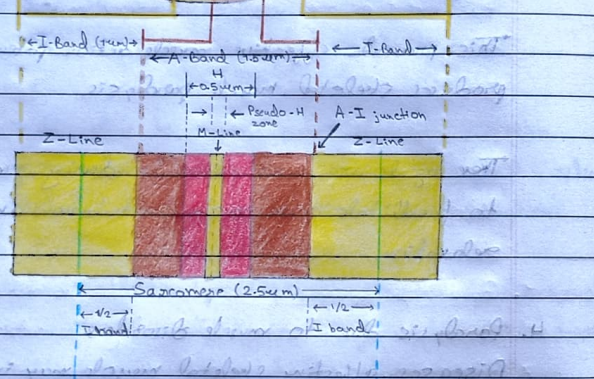
Paralysis results from failure of normal motor activity. Its physiological basis may involve nerve conduction, neuromuscular transmission, or skeletal muscle contraction. Understanding these mechanisms helps explain neuromuscular disorders.

Sarcomere

(A) General appearance



(B) light microscopic appearance



(C) Electron microscope appearance

